

MUHAMMAD ZAIN AKRAM

Full-Stack Web Developer | React.js · Next.js · Express.js · Node.js · AI Integration

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[LinkedIn](#) | [GitHub](#) | [Anthropic Courses](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

My journey in web development is powered by an array of unique tools and languages, with JavaScript at the core of everything I build. With 2+ years of experience, I work closely with frameworks like React.js and Next.js to craft seamless, user-friendly websites. And strong design skills, I focus on building fast, secure, and visually striking web experiences. Alongside clean frontend development, I am also skilled in training Machine Learning models. I love exploring new tech and am eager to bring my skillset to a team, learn from experienced developers, and contribute to real-world projects.

Front-end: React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, Tailwind CSS, Bootstrap, Material UI, Framer Motion, Three.js

Backend: Node.js, Express.js

Database & APIs: PostgreSQL, MongoDB, SQLite, JWT Authentication, SMTP APIs

ML Libraries: Scikit-learn, Pandas, NumPy, Matplotlib, OCR, Time-Series Forecasting.

ML Models: Gradient Boost, LSTM, RNN, CNN, Linear Regression, Random Forest etc.

DevOps & Tools: Docker, Git, GitHub, Vercel, Agile methodology, Netlify

PROJECT EXPERIENCE

Pak Booking Hotel Booking Platform (Professional Project)

Sep 2025

Django REST Framework · Next.js · PostgreSQL · Docker · JWT · Stripe

- Architected a production-ready full-stack hotel booking platform for Pakistan's hospitality sector with JWT authentication and role-based access control.
- Engineered complete booking lifecycle (Pending → Confirmed → Completed → Refunded) with automated email notifications and real-time status updates.
- Integrated Stripe/Razor Pay payment gateways, configurable refund processing, and an admin dashboard with financial analytics and reporting.
- Achieved Lighthouse score of 90+ through PostgreSQL query optimization, Docker containerization, and PWA with offline support.
- Implemented multi-language support (English, Urdu, Roman Urdu) with advanced search filters and fully responsive UI.

AI-Based Smart Energy Consumption Prediction System (FYP)

March 2025 – Aug 2025

Python · Django · React.js · Tailwind CSS · ARIMA/LSTM · OCR · Google Gemini · RAG | Final Year Project

- Built an end-to-end AI web application predicting next-month energy bills from historical usage data with personalized saving recommendations.
- Implemented ARIMA/LSTM time-series forecasting models projecting energy usage and billing trends through 2030.

- Integrated an OCR module to automatically extract consumption data from uploaded utility bill images, eliminating manual data entry.
- Developed SECPARS, a context-aware chatbot using Google Gemini with lightweight RAG, delivering 24-hour energy forecasts via Django REST and React + Tailwind UI.

Online Banking System (University Project)

Jun 2025

MERN Stack: MongoDB · Express.js · React.js · Node.js

- Designed and developed a secure MERN-stack banking application enabling account creation, fund transfers, bill payments, and real-time transaction tracking.
- Implemented RESTful APIs with JWT authentication, and role-based dashboards for users and admins.

EDUCATION

B.S. Computer Science

2021 – 2025

Lahore Garrison University, Lahore, Pakistan

Relevant Coursework: Data Structures & Algorithms, OOP, Database Management Systems, Computer Networks, Machine Learning, Deep Learning, Web Development & Design, Compiler Construction.

Intermediate (Pre-Engineering)

2018 – 2020

Al-Zahrawi Science College, Lahore

Matriculation

2016 – 2018

Bright Scholar, Lahore

CERTIFICATIONS & TRAINING

- Claude 101 Certificate of Completion, Anthropic (2025). Credential: anthropic.skilljar.com. Covered fundamentals of working with Claude AI, prompt design, and responsible AI usage in real-world applications.
- AI Fluency: Framework & Foundations Certificate of Completion, Anthropic in partnership with University College Cork, Ringling College of Art + Design, HEA & National Forum (2025). Credential: anthropic.skilljar.com. Covered AI frameworks, foundational concepts, and practical application of generative AI tools.
- LEGO Robotics & Engineering Design Workshop GEARS Society, Lahore Garrison University (Jan 2022). Covered LEGO-based prototyping, robot programming, logical reasoning, and engineering design concepts.
- Here is the drive link:
https://drive.google.com/file/d/1muQGMhYHnAwFHIImqeYPBp5vdjkr4W_0/view?usp=sharing